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Letter

Physics-informed neural network model for transient thermal analysis of superconductors

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Abstract

Thermal stresses in superconducting tapes, resulting from quench, over-current conditions, or high-current faults, can rapidly elevate temperatures, potentially causing hot spots, performance degradation, localized damage, or catastrophic burnout failure if heat dissipation is inadequately managed. Traditional numerical approaches, such as finite element analysis (FEA), accurately simulate these transient thermal behaviors but suffer from significant computational burdens, complex meshing processes, and sensitivity to input and meshing parameters. Moreover, FEA results often lack generalization capability, necessitating new simulations for each variation in problem parameters. Recently, artificial intelligence (AI) and machine learning (ML) methods have emerged as alternatives for addressing superconductivity problems. However, existing AI/ML models typically operate as black boxes, providing little insight into the physics-based relationships between inputs and outputs. To address this limitation, this study introduces a physics-informed neural network (PINN) framework specifically designed to model transient thermal behaviors in superconducting tapes. By embedding governing physics laws directly into the neural network's training process, PINNs offer increased transparency, accuracy, and interpretability compared to conventional AI/ML models. This paper, for the first time, aims to demonstrate PINNs as an innovative modeling approach for superconductors, focusing explicitly on transient thermal responses following a heat pulse in superconducting tapes. Validation against analytical solutions and benchmark FEA simulations confirms the PINN model's effectiveness in accurately capturing sharp thermal gradients and transient heat transfer. Furthermore, results highlight the model's excellent generalization and extrapolation

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capabilities, illustrating that a trained PINN can reliably predict scenarios beyond its original training dataset, thereby offering significant potential for tackling complex challenges related to superconductors.

Keywords: artificial intelligence, hottest point, machine learning, normal-zone propagation, physics-informed modeling, PINN

1. Introduction

By offering negligible electrical resistance in their superconducting state, coated conductor tapes enable highly efficient energy transfer in power systems and the generation of powerful magnetic fields for fusion reactors [1]. However, one of the critical challenges for their practical deployment is managing their thermal behavior under dynamic transitions and heat pulses, e.g. quench, over-current, or large electric faults [2]. These transitions often lead to rapid, localized heating that can propagate throughout the conductor, which can induce significant thermal and electrical stresses on the superconducting layers in the tape. If these stresses are not effectively managed, they can severely damage or even destroy the superconducting layer/tape [3, 4]. The capability to accurately model transient thermal events is paramount for taking mitigation strategies, such as quench detection circuits, active cooling, fault tolerance, and current diversion paths [5].

Traditional finite element methods (FEMs) and finite difference methods (FDMs) have long served as the standard approaches for simulating thermal transients in superconductors [6–8]. These techniques solve partial differential equations (PDEs) by discretizing the governing equations and the geometry, and performing iterative calculations that trade off efficiency and accuracy. However, high-fidelity simulations of the aforementioned thermal transients are often computationally expensive, particularly for large-scale power devices with complex geometries, multi-scale physics, or real-time control scenarios where rapid predictions are essential. Also, results of FEM and FDM cannot be generalizable for different transient scenarios, and thus, the simulations need to be run again (from scratch) for any new conditions [9].

Recent advances in artificial intelligence (AI) and machine learning (ML) have prompted the development of more intelligent strategies, but they depend on the availability of data [10–12]. Physics-informed neural networks (PINNs), as a new intelligent modeling technique, embed the governing physics equations (shown in figure 1) directly into the neural network's loss function, effectively constraining the network to abide by known laws of physics [13]. In this way, instead of being treated as a 'black box', the network learns solutions that inherently respect governing laws, boundary conditions (BCs), and the initial conditions (ICs) of the problem. Therefore, as being shown in figure 2, PINNs can be

trained without the existence of any experimentally or numerically sourced data [14] and show high accuracy where no data is available. Another key advantage is that PINNs are a 'mesh-free' technique, requiring only randomly distributed points within the domain during training and thus eliminating the need for elaborate mesh generation—a crucial distinction from FEM or FDM-based methods, which are highly sensitive to mesh quality and resolution. This mesh-free characteristic also grants remarkable flexibility and adaptability, enabling PINNs to handle complicated or evolving geometries with minimal reconfiguration [15]. Additionally, boundaries and ICs are naturally integrated into the loss function, streamlining the treatment of complex physics constraints that might otherwise require specialized implementations in classical PDE solvers. Moreover, as the solution is directly represented by a continuous neural network, PINNs provide immediate access to both the solution field and its derivatives at any point in the domain, which is particularly useful for sensitivity analyses and design optimization. PINNs also scale effectively to more complex or higher-dimensional problems, leveraging parallel GPU computation to expedite training times and facilitate real-time or near-real-time predictions [16]. Moreover, PINNs can be extended to solve inverse problems by incorporating observed data into the loss function—allowing unknown parameters, BCs, or source terms to be evaluated alongside the primary field solution [17, 18]. These strengths make PINNs a highly promising framework for modeling complex superconductor problems, where computational efficiency, adaptability, and strict adherence to physics laws are paramount.

For the first time, this paper applies PINN as a promising modeling technique for a superconductivity problem, focusing on transient thermal analysis of superconducting tapes, which opens new opportunities for rapid, accurate predictions of temperature distributions as well as showcasing the capacity of using PINNs for more complex superconductor problems.

In the following, the physics of the problem and the assumptions, along with the PINN's architecture and the way it incorporates PDEs into the training process, are explained in section 2. Then, the results of the PINN are validated with the values of the analytical technique and FEM model in COMSOL Multiphysics [19] for the same problem in section 3. In section 4, a conclusion is made based on the results of the work, and the potential use of the PINN for future work on superconducting problems is discussed.

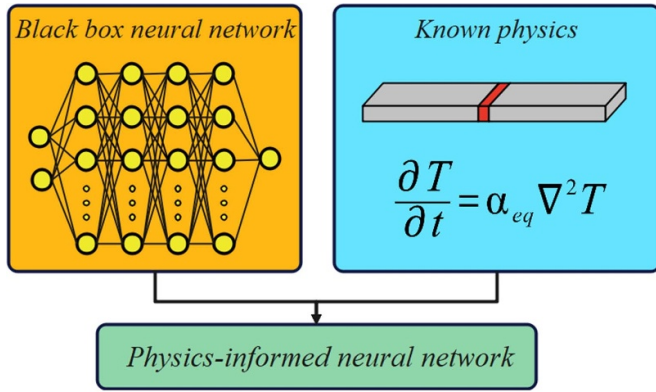


Figure 1. The concept of developing a PINN framework by embedding physics constraints and equations into neural networks.

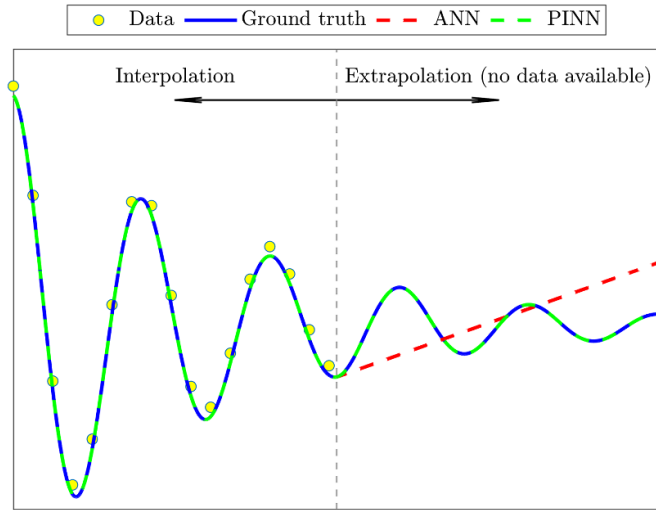


Figure 2. A schematical difference between data-driven ANN and PINN.

2. PINN for thermal analysis of superconductors

In order to do the thermal analysis, a BSCCO tape is modeled with 1 m length, 0.001 m width (reflecting a narrow strip for simplified analysis), and 0.00021 m thickness [20]. To simplify the problem, the tape is considered as a homogeneous material and equivalent thermal properties, including different layers of the tape (copper, HTS layer, etc.), were calculated according to the described method in. Also, the influence of the convectional and radiational heat transfer modes was excluded, and the problem was considered as conduction only.

To evaluate the temperature distribution and thermal analysis of the tape, the transient heat transfer equation needs to be solved over the defined space and time domain. The general form of this equation, considering no internal or external heat sources, can be written as:

Table 1. Properties of the simulated tape.

Property	Value
Type of tape	BSCCO
Length	1 [m]
Width	0.001 [m]
Thickness	0.00021 [m]
Base temperature	77 [K]
k_{eq}	150 [W m ⁻¹ · K]
ρ_{eq}	9000 [kg m ⁻³]
C_{peq}	217 [J kg ⁻¹ · K]
α_{eq}	7.68×10^{-5} [m ² s ⁻¹]

$$\frac{\partial T}{\partial t} = \alpha_{eq} \nabla^2 T = \alpha_{eq} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) \quad (1)$$

where T is the temperature, t is time, x, y, z are space variables, and α_{eq} is the equivalent thermal diffusivity of the superconducting tape which can be evaluated from the equation below:

$$\alpha_{eq} = \frac{k_{eq}}{\rho_{eq} C_{peq}} \quad (2)$$

where k_{eq} , ρ_{eq} , and C_{peq} are equivalent thermal conductivity, density, and specific heat of the tape. Referring to [20], these properties are summarized in table 1. While these properties are generally a function of temperature, over the temperature range considered in this study (77–300 K) the density ρ_{eq} is effectively constant, and prior studies have shown that treating k_{eq} and C_{peq} as constant for BSCCO tapes does not materially affect the predicted temperature profile [20–25]. Hence, the effect of temperature on α_{eq} was taken out of consideration which reduced the time of simulation by an order of magnitude by introducing minor errors to the results.

As the thickness and the width of the tape are too small compared with the length of the tape, the heat distribution in y and z directions are uniform and can be neglected. This assumption was also verified by developing a 3D FEM model of the tape in COMSOL and comparing its results with a 1D version of it. This occurs because the high in-plane thermal conductivity of the copper stabilizer rapidly equilibrates the tape's small cross-section, so axial conduction dominates the early-time response. Therefore, equation (1) can be rewritten as below:

$$\frac{\partial T}{\partial t} = \alpha_{eq} \frac{\partial^2 T}{\partial x^2}. \quad (3)$$

To solve this equation, a PINN framework is adopted. In this approach, the neural network is trained to approximate the solution to the PDE by incorporating the physics directly into its loss function. A feed-forward neural network (FFNN) serves as the underlying architecture for the PINN. An FFNN is a fundamental type of artificial neural network characterized by the unidirectional flow of information from input to output

layers without any cyclic or recurrent connections [26]. It consists of several layers of interconnected nodes, systematically organized into an input layer, one or more hidden layers, and an output layer. Each neuron within these layers receives input signals, processes them through a weighted sum followed by a nonlinear activation function that introduces non-linearity into the network, and transmits the resulting signal to subsequent layers [27].

Unlike conventional ANNs, which minimize only the mean-squared error (MSE) of the training data and are therefore highly sensitive to network size, a PINN augments the MSE with physics-based residuals. These additional constraints allow the network to learn complex, non-linear relationships even with small architectures.

To confirm this, we trained networks with two to six hidden layers. The two- and three-layer models could not reproduce the Gaussian temperature profile, but architectures with four or more layers achieved relative errors below 0.2%. It was found that increasing the depth beyond four layers, however, drastically raised training time (simulation time) and RAM usage, which hindered further exploration of larger networks due to hardware limitations. Also, as was discussed in [28], proceeding with relatively small networks that can satisfy the expected accuracy from the simulation, alleviates the vanishing gradient problem typically encountered in deep networks.

Overall, we tested eight configurations—[60 60], [60 60 60], [30 30 30 30], [60 60 60 60], [120 120 120 120], [30 30 30 30 30], [60 60 60 60 60], and [60 60 60 60 60 60]. Except for the first two, all produced accurate results, but the four-layer [30 30 30 30] network offered the best trade-off between accuracy and computational cost. Consequently, we report our findings using this architecture.

The activation function used between the layers is the hyperbolic tangent function ($\tanh$), which is a bounded (from -1 to 1) and smooth activation function [29], properties that are well suited for this work where the temperature profile after quench has sharp changes near the hot zone (shown in figure 5). If an unbounded activation (e.g. ReLU) is used, small errors in the hot zone can be amplified as the network tries to reconstruct the sharp central peak. Also, unlike logistic functions, $\tanh$ is zero-centered, i.e. positive and negative inputs have symmetric derivatives. This reduces the bias-shift effect during back-propagation (which is critical for training of PINNs) and helps gradients propagate deeper, reducing both vanishing and exploding-gradient incidents. Figure 4 visualizes these functions for an input between -11 and 11 to show how different their results are.

During the operation of an FFNN, data is introduced at the input layer, where neurons directly represent features of the input vector. These signals propagate sequentially through hidden layers, where internal representations of the input data are progressively constructed. This layered architecture enables FFNNs to capture complex, nonlinear relationships [30].

Training involves optimizing its weights and biases, typically using gradient-based methods to minimize the loss function. In this study, we utilize the Adam optimizer [31] (Adaptive Moment Estimation), a widely used algorithm in

deep learning that combines the advantages of AdaGrad [32] and RMSProp [33]. Adam optimizer maintains per-parameter learning rates, which are adapted based on estimates of the first and second moments of the gradients, facilitating faster and more robust convergence.

The backpropagation algorithm is used to calculate the loss function's gradient with respect to weight and bias factors in the network by propagating the error backward from the output layer. Through iterative updates, the model gradually improves its performance, enhancing both accuracy and generalization to unseen data.

Through the loss function, the physics of the problem can be introduced to the network to construct the PINN framework, as being illustrated in figure 3. The loss function is defined as follows:

$$\zeta_{\text{total}} = \zeta_{\text{IC}} + \zeta_{\text{BC}} + \zeta_{\text{PDE}} \quad (4)$$

where ζ_{total} is the loss function that is used for network optimization. The ζ_{BC} , ζ_{IC} and ζ_{PDE} are the losses of BC, IC, and heat equations, respectively. Each of these loss components introduces the prediction inaccuracies of the network with the physics of the problem.

By transferring the sides of equation (3) to one side and defining u as the evaluated temperature by PINN, we have:

$$\zeta_{\text{PDE}} = \frac{\partial u}{\partial t} - \alpha_{\text{eq}} \frac{\partial^2 u}{\partial x^2}. \quad (5)$$

Therefore, the first-order time derivative and second-order space derivatives of u are required, which can be evaluated through automatic differentiation techniques that neural networks offer. Auto-differentiation is a technique that automatically calculates exact derivatives of functions modeled by neural networks by tracking operations during the forward pass and then applying the chain rule in reverse. In PINNs, it enables the neural network to directly compute partial derivatives of its outputs with respect to the input variables, ensuring that the governing physics equations and BCs are embedded in the training process.

As the heat pulses caused by thermal transients such as quench, over-current, and high current faults often result in a very high temperature in a very small zone of tape, we define the initial temperature in the form of a Gaussian function of the length, which is a common practice in the literature [34–37]. Gaussian function is differentiable, which ensures smooth first- and second-order derivatives that are required in solving the heat transfer equations. This avoids the oscillations that might be caused by top-hat or triangular pulses and stabilizes the automatic-differentiation step during PINN training, as well as helping the FEM solver to converge more easily. Therefore, the initial temperature is defined as equation (6) and has been shown in figure 5:

$$T(x, 0) = T_{\text{base}} + \Delta T * \exp\left(-A * \left(x - \frac{L}{2}\right)^2\right) \quad (6)$$

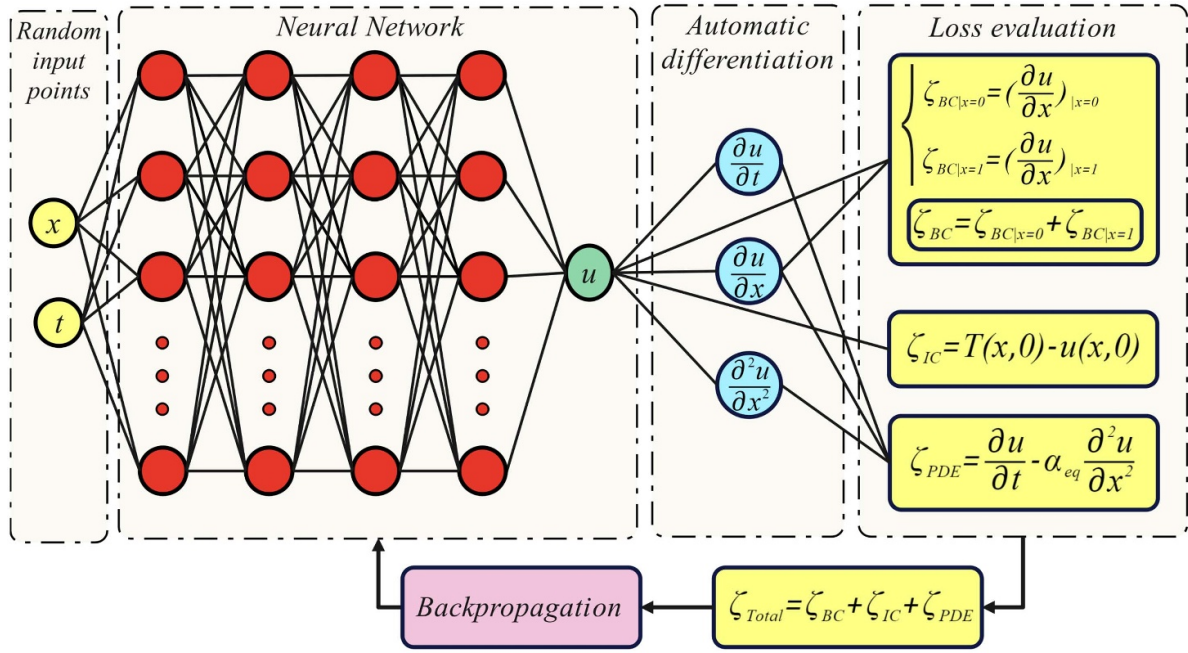


Figure 3. Schematic of the process of developing PINN.

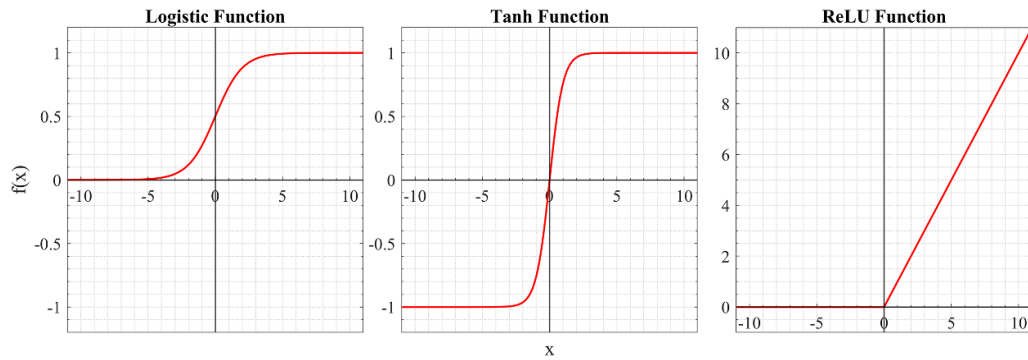


Figure 4. Visualization of differences between logistic function (left), Tanh function (middle), and ReLU function (right).

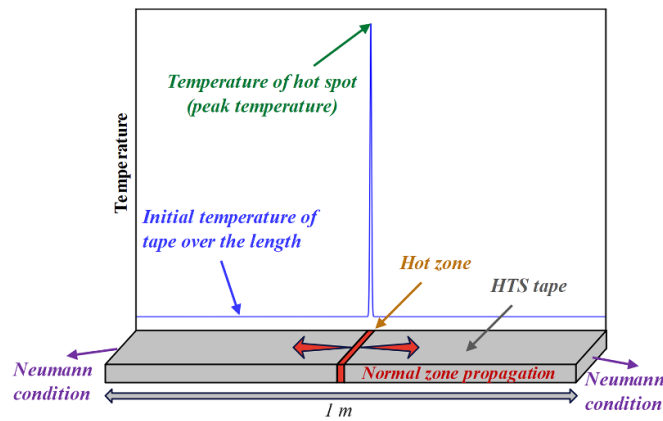


Figure 5. Defined initial temperature for the tape.

where T_{base} is the base temperature of the tape before the transient event, which is 77 K, ΔT is the temperature difference that has been caused by a transient event, A is a constant to control the sharpness of the initial temperature profile equal to 1.8×10^5 , and L is the length of the HTS tape. By changing ΔT , different hot spot (peak) temperatures can be modeled using the PINN model. The value of A was chosen based on a sensitivity analysis of values from 1×10^3 – 5×10^5 where the aim was to have a close match of the temperature profile to the reported values in the literature. Using equation (6), the loss function for IC can be written as follows:

$$\zeta_{\text{IC}} = u(x, 0) - T(x, 0). \quad (7)$$

Because the length of the tape is far longer than the length of the hot zone, the infinite length condition can be applied to the modeling of the tape. Therefore, the Neumann BCs were applied to both ends of the tape, which represents adiabatic/no heat flux conditions:

$$\begin{cases} \left(\frac{\partial T}{\partial x} \right)_{|x=0} = 0 \\ \left(\frac{\partial T}{\partial x} \right)_{|x=1} = 0 \end{cases}. \quad (8)$$

Similarly, the loss function for BCs can be defined as:

$$\begin{cases} \zeta_{\text{BC}|x=0} = \left(\frac{\partial u}{\partial x} \right)_{|x=0} - \left(\frac{\partial T}{\partial x} \right)_{|x=0} = \left(\frac{\partial u}{\partial x} \right)_{|x=0} \\ \zeta_{\text{BC}|x=1} = \left(\frac{\partial u}{\partial x} \right)_{|x=1} - \left(\frac{\partial T}{\partial x} \right)_{|x=1} = \left(\frac{\partial u}{\partial x} \right)_{|x=1} \end{cases} \quad (9)$$

$$\zeta_{\text{BC}} = \zeta_{\text{BC}|x=0} + \zeta_{\text{BC}|x=1}. \quad (10)$$

In order to calculate these losses and train the PINN, a set of spatiotemporal points is required. Unlike traditional numerical methods such as FEM, which rely on a structured mesh to discretize the domain into elements and nodes, PINNs utilize randomly or strategically distributed collocation points throughout the domain and its boundaries. The primary aim of collocation point selection is to ensure comprehensive domain coverage, particularly emphasizing areas near boundaries and regions exhibiting sharp gradients, where the solution changes rapidly. Initially, a uniform random distribution of collocation points within the domain may be adopted.

3. Results and analysis

3.1. PINN training

To effectively capture the dynamics near boundaries and steep gradients in the middle of the tape, additional random points were generated to provide a denser distribution of training points in the middle area. The generated sampling points are shown in figure 6.

The neural network is now ready to be trained using the total loss function and the Adam optimizer. During training, Adam adjusts the weights and biases of the network's neurons to minimize the total loss. A convergence threshold of 10^{-5} is set, meaning the training process stops once the total loss falls below this value.

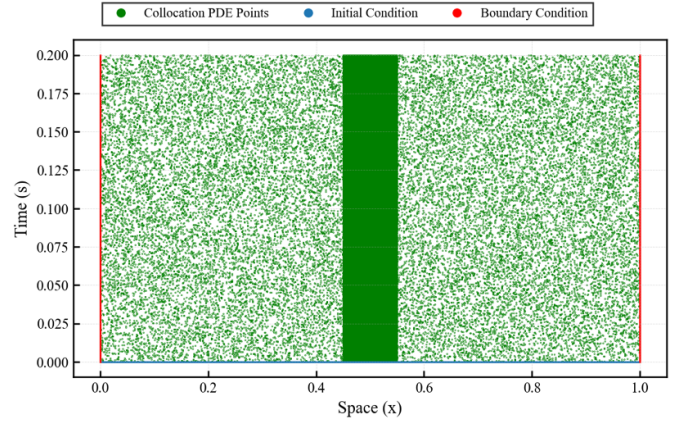


Figure 6. Random datapoint generation for training of the neural network.

Figure 7 illustrates the evolution of the different loss components separately—IC, BC, and PDE—along with the total losses throughout the training of the PINN. As shown, the total loss (green line) decreases rapidly during the initial stages and gradually converges towards the predefined threshold of 10^{-5} , indicating successful learning of the underlying physics. Among the loss components, the PDE loss (yellow dashed line) dominates for most of the training process, while the IC and BC losses (blue and red lines, respectively) remain relatively small. In the first 20 000 epochs—where an epoch refers to one complete pass through the entire training dataset in an ANN—the PDE loss shows a significant decline, whereas the IC and BC losses fluctuate, with a generally increasing trend. The convergence behavior of all individual loss components confirms that the neural network effectively learns and satisfies the initial, boundary, and governing equation constraints of the problem.

3.2. Validation

Once the PINN is well trained in the physics of the problem, it can be used for accurate evaluation of the temperature for any given time of simulation and position of the tape. In this section, the performance of PINN is examined for interpolation tasks, where the results of the simulation are required within the trained period of time (0–0.2 s).

In order to provide a baseline model for validation of the results, an analytical solution of this transient thermal analysis problem was used as the base model in which the heat equation has been mathematically solved using the defined IC and BC constraints:

$$T(x, t) = T_{\text{base}} + \frac{\Delta T}{\sqrt{1 + 4\alpha t A}} \cdot \exp\left(-\frac{A(x - \frac{L}{2})^2}{1 + 4\alpha t A}\right). \quad (11)$$

Figure 9 illustrates the temperature profile evaluated by the PINN model (solid lines) and the analytical model (dashed black line) at selected time points (0.00 s, 0.07 s, 0.13 s, and 0.20 s), clearly capturing the transient nature of the heat transfer. Each subplot (a–d) represents scenarios with distinct initial peak temperatures: (a) 150 K, (b) 200 K, (c) 250 K, and

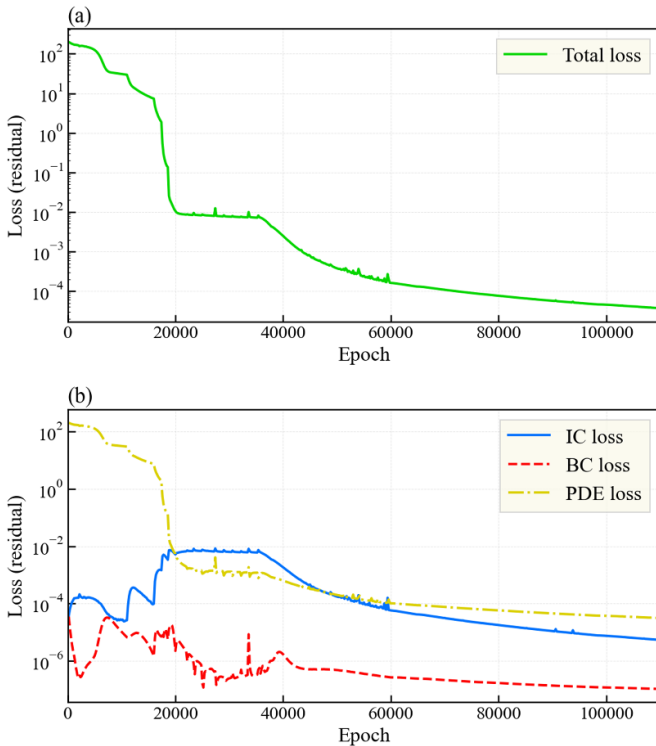


Figure 7. Convergence of different elements of losses during PINN optimization process (a) total losses (b) other elements.

(d) 300 K. Initially ($t = 0.00$ s), the temperature peaks sharply at the center position (0.50 m), reflecting the prescribed initial Gaussian temperature distribution. As time goes by ($t = 0.07$ s, 0.13 s, and 0.20 s), the temperature distribution broadens and flattens due to heat conduction, consistently observed across all scenarios. Notably, the PINN predictions align remarkably well with the analytical model, accurately reproducing both the peak magnitudes and the gradual diffusion of heat.

Figure 8 provides a comprehensive visual comparison between the temperature distribution evaluated by the PINN model (left subplots) and the corresponding analytical model (middle subplots), alongside the pointwise absolute differences between these two sets of results (right subplots). Each row represents a scenario with different initial peak temperatures at the hot spot—specifically 150 K, 200 K, 250 K, and 300 K from the top to bottom, respectively.

Across the time range (0–0.2 s) and the selected spatial domain (0.48–0.52 m), the PINN-generated temperature distributions exhibit excellent agreement with the analytical model, capturing the temporal evolution and spatial propagation of heat accurately.

Detailed examination of the error plots (right subplots) reveals that the absolute errors between the PINN and analytical solutions are consistently below 1% relative error throughout the evaluated range. Errors are predominantly localized near the central region (at approximately 0.50 m), where the initial temperature peak defined by the Gaussian distribution occurs. Notably, errors slightly increase with higher initial

temperatures; however, even in the worst-case scenario at an initial temperature of 300 K, the error remains small, underscoring the robustness of the PINN method in approximating complex thermal analysis.

Additionally, at all initial temperatures, the errors are transient, which peak at early time stages (approximately within the first 0.05 s) and diminish subsequently, which highlights the PINN's capability to quickly stabilize and provide reliable results. This trend is consistent across all four cases, emphasizing that the model error does not propagate significantly with time, thus validating the effectiveness of the PINN approach in simulating heat transfer scenarios with varying ICs.

3.3. Extrapolation capabilities: beyond simulation time

Extrapolation is a crucial aspect of each modeling technique in scientific and engineering problems, particularly for superconducting applications, as it enables us to predict system behavior beyond the range of available data and simulated conditions. While interpolation deals with known data points, extrapolation tests a model's generalization capability and robustness, revealing its ability to handle unseen scenarios.

To compare the extrapolation capacity of PINN with other techniques, a FEM model has also been developed in COMSOL Multiphysics v6.3 to make a comprehensive comparison. To do so, using the 'Heat transfer in Solids' module of COMSOL, the initial temperature was set according to equation (6). Also, zero heat flux BCs were applied to the tape according to equation (8). The material properties of the HTS tape were also defined based on the details in table 1. Due to sharp changes in the hot zone, the physics-controlled mesh was unable to capture the initial temperature. Therefore, to ensure the model captures the initial temperature and sharp changes in the hot zone, a mapped mesh with minimum element size of 0.0002 was used. The mesh was then redistributed with exponential growth and element ratio of 0.0001 to make sure the hot zone had enough mesh elements to capture all changes. Finally, a time-dependent solver with a time step of 0.0001 s was used to simulate the transient temperature profile of the tape.

Focusing only on the temperature variations of the hot spot, as shown in figure 10, while both PINN and COMSOL simulations are trained or run from 0 to 0.2 s, yet their extrapolation behavior beyond this range (up to 0.5 s) reveals a clear difference in performance. Within the trained interval, all three approaches—PINN, COMSOL, and Analytical models—align closely. However, beyond 0.2 s, the COMSOL model results start to deviate more significantly from the analytical solution, especially at lower initial temperatures. In contrast, the PINN model continues to closely follow the analytical trend, suggesting it has learnt the underlying physics more effectively and is capable of better generalization.

Table 2 also presents a comparative evaluation of the PINN and COMSOL simulation results against the analytical (exact) solution for both interpolation and extrapolation scenarios across different peak temperature values (150 K to 300 K)

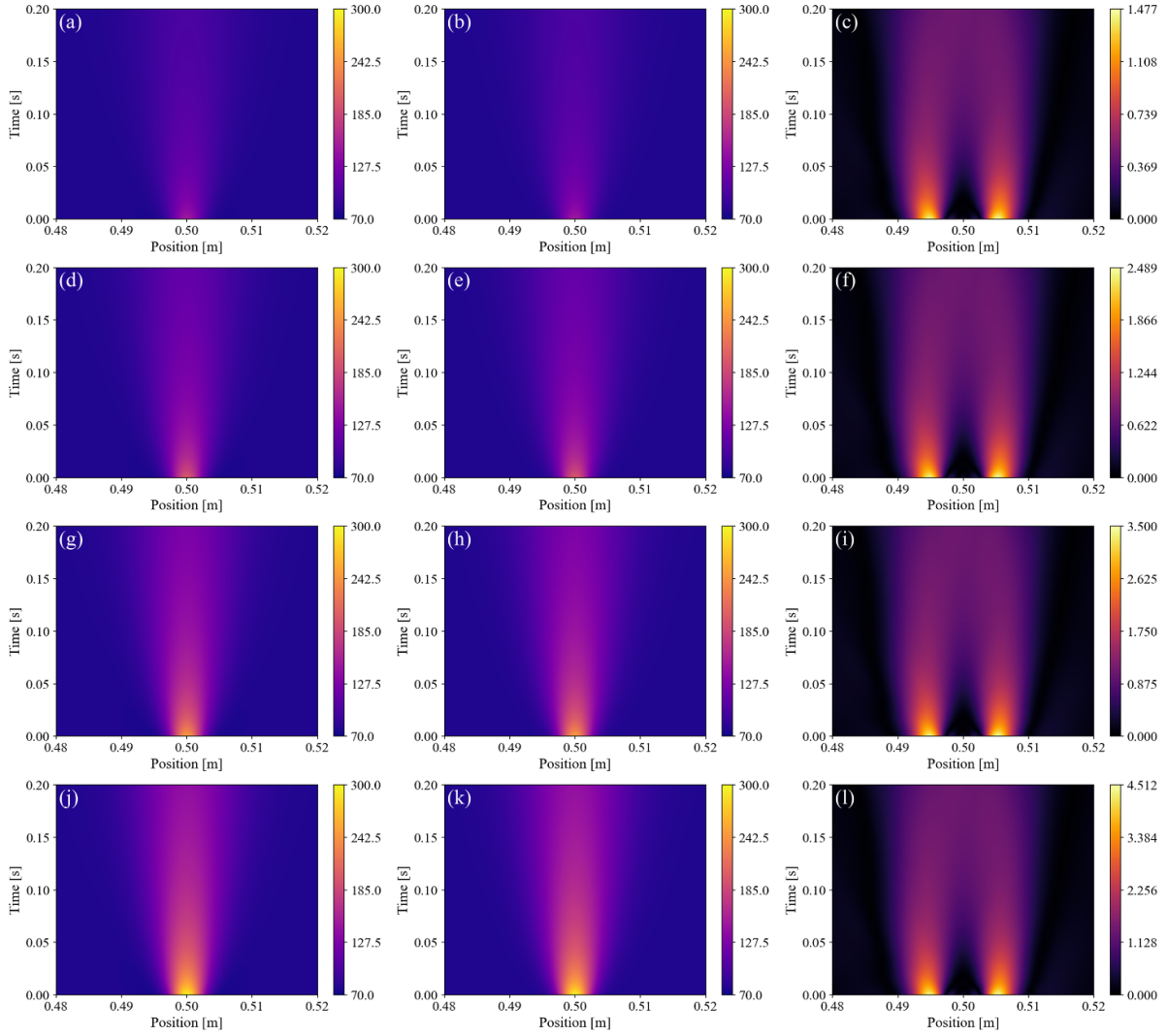


Figure 8. Comparison between the evaluated temperature profiles obtained by PINN (left column: (a), (d), (g), (j)) and analytical model (middle column: (b), (e), (h), (k)), along with the pointwise absolute error between these models (right column: (c), (f), (i), (l)) for different initial hot-point temperatures.

using standard performance metrics. The statistical metrics are defined as follows:

$$R^2 = 1 - \frac{\sum_{k=1}^K (u_k - \hat{u}_k)^2}{\sum_{k=1}^K (u_k - \bar{u})^2} \quad (12)$$

$$\text{MAE} = \frac{1}{K} \sum_{k=1}^K (u_k - \hat{u}_k) \quad (13)$$

$$\text{MARE} = \frac{1}{K} \sum_{k=1}^K \frac{(u_k - \hat{u}_k)}{u_k} \quad (14)$$

where in these equations, we considered u as the actual value, $\bar{u}$ as the average of u for K data points, $\hat{u}$ as the estimated value of the critical current, and K for the number of data points.

As can be seen with this table, in the interpolation range, both PINN and COMSOL exhibit excellent agreement with

the analytical solution, with R^2 consistently exceeding 0.999 for both methods. However, COMSOL slightly outperforms PINN in terms of error metrics, achieving lower mean absolute error (MAE) and mean absolute relative error (MARE). For instance, at 300 K, COMSOL achieves an MAE of 0.2447 and a MARE of 0.1453%, compared to PINN's MAE of 0.7253 and MARE of 0.5739%.

In extrapolation, however, the performance gap widens in favor of the PINN. While PINN maintains a high R^2 value of 0.9957 across all temperatures and shows a modest increase in error with rising peak temperature, COMSOL's accuracy drops significantly. At 300 K, COMSOL's R^2 drops to 0.9137 and MARE increases sharply to 3.41%, whereas PINN sustains a notably lower MARE of 0.9659%. This indicates that while COMSOL is highly accurate within the range of training or known conditions, its extrapolation capability is limited. In contrast, PINN demonstrates strong generalization performance in both interpolation and extrapolation tasks (beyond the

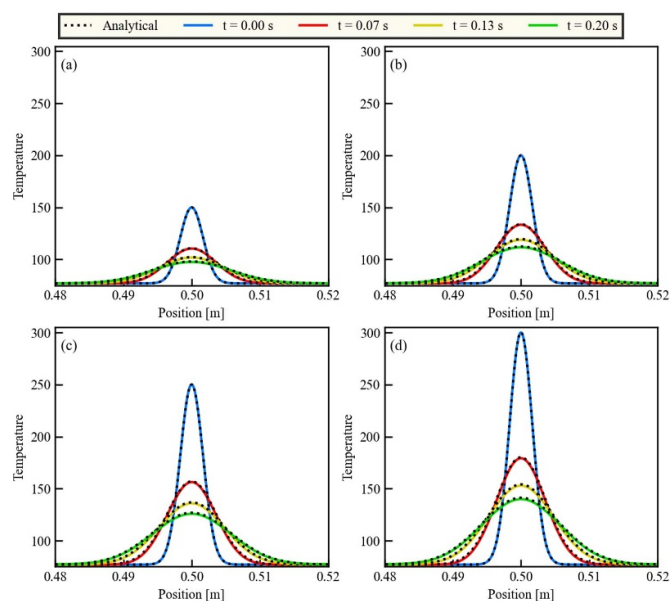


Figure 9. The temperature variations over the length of the tape at different simulation times.

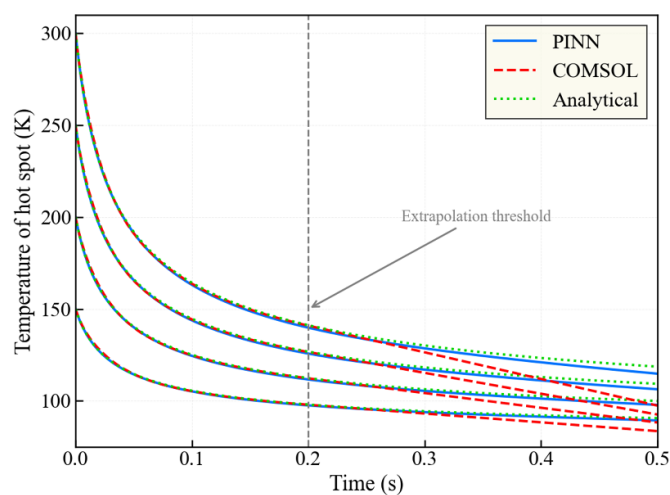


Figure 10. The temperature variations of the hot spot over time under different initial temperatures.

Table 2. Comparison of the accuracy of PINN and COMSOL models in comparison to the analytical base model.

		Interpolation			Extrapolation		
		R2	MAE	MARE (%)	R2	MAE	MARE (%)
PINN	150	0.9996	0.2374	0.2496	0.9957	0.3758	0.4125
	200	0.9996	0.4001	0.3764	0.9957	0.6331	0.6299
	250	0.9996	0.5627	0.4824	0.9957	0.8905	0.8114
	300	0.9996	0.7253	0.5739	0.9957	1.1479	0.9659
COMSOL	150	0.9998	0.1114	0.1013	0.9101	1.3638	1.4969
	200	0.9999	0.1565	0.1197	0.9135	2.251	2.2361
	250	0.9999	0.1988	0.1326	0.9105	3.2241	2.925
	300	0.9999	0.2447	0.1453	0.9137	4.0777	3.4079

training domain), making it a promising tool for predictive modeling in scenarios where data is sparse or extrapolation is required.

4. Conclusion and future works

This paper, for the first time in the superconducting community, has demonstrated the efficacy and robustness of PINNs in modeling the transient thermal behavior of superconducting tapes during heat pulse conditions. PINNs have successfully addressed several limitations inherent to traditional FEM-based approaches, notably overcoming mesh dependency and significantly reducing computational requirements. By embedding governing physics directly into the neural network structure, PINNs eliminate iterative mesh refinement, achieving greater efficiency and adaptability. Also, by training a network that incorporates the physics constraints of the superconducting problem, PINN has shown great potential for extrapolation tasks where it can predict beyond the training range with high accuracy.

Despite these promising results, several areas remain open for exploration and development to further enhance PINN-based methodologies:

Currently, complex PINN simulations require extensive training epochs, contributing to longer simulation times. Future research should explore advanced computational techniques such as parallel and distributed computing environments and GPU-accelerated hardware to substantially reduce training durations. Leveraging modern GPU platforms and optimization frameworks can potentially enable near-real-time predictions necessary for active control and protection systems in superconductors.

The present model simplifies certain physics complexities, considering homogenous materials and conduction-only heat transfer. Future developments should focus on incorporating multi-physics phenomena, including electromagnetic interactions, mechanical stresses, and combined convective-conductive heat transfer effects. Such comprehensive models will better reflect real-world operational scenarios of superconducting devices and significantly improve predictive accuracy and applicability.

The proposed PINN architecture utilizes a FFNN. Investigating deeper network structures like convolutional neural networks (CNNs) could enhance the network's capability to capture intricate spatial-temporal features, particularly in scenarios involving more complex geometries and BCs. Further, optimization algorithms beyond the Adam optimizer, including second-order optimizers or hybrid methods, could improve training convergence and stability.

Extending the demonstrated 1D thermal modeling capability to multi-dimensional scenarios is essential for addressing realistic engineering problems. Future efforts should involve developing PINNs capable of handling 2D and 3D geometries typical of superconducting devices, considering anisotropic and non-uniform thermal properties across different material layers, and complex BCs arising from realistic operational environments.

It is also worth mentioning that an exciting avenue for PINNs is their application to inverse problems, allowing simultaneous prediction and identification of unknown parameters, such as material properties, defect locations, or thermal and electromagnetic source terms. Future research should emphasize developing robust inverse methodologies within the PINN framework, enabling precise diagnostics and predictive maintenance of superconducting systems.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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